Homework 5

Math 742

2/30/2026

Problem 1 — Consistency (C&B Ch 5.5)

Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Exponential}(\lambda)$, where $\lambda > 0$.

Define the estimator
$$\hat{\lambda}_n = \frac{1}{\bar{X}_n}.$$

a) Show that $\bar{X}_n \xrightarrow{p} 1/\lambda$.

b) Use a continuous mapping argument to prove that $\hat{\lambda}_n$ is a consistent estimator of λ.

c) Explain intuitively why consistency holds even though $1/\bar{X}_n$ is nonlinear.

Problem 2 — CLT and Slutsky (Wasserman Ch 5)

Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Bernoulli}(p)$.

Define $\hat{p} = \bar{X}_n$.

a) Use the Central Limit Theorem to show

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p)).$$

b) Show that

$$\frac{\sqrt{n}(\hat{p} - p)}{\sqrt{\hat{p}(1-\hat{p})}} \xrightarrow{d} N(0, 1).$$

(Hint: Apply Slutsky's theorem.)

c) Explain why this result justifies the large-sample Wald confidence interval for p.

Problem 3 — Delta Method (Wasserman Ch 9.9)

Let $X_1, \dots, X_n \overset{iid}{\sim} \text{Poisson}(\lambda)$.

We know $\hat{\lambda} = \bar{X}_n$.

We are interested in estimating
$$\theta = \log(\lambda).$$

Define $\hat{\theta} = \log(\bar{X}_n)$.

a) Use the CLT to derive the asymptotic distribution of $\bar{X}_n$.

b) Apply the Delta Method to find the asymptotic distribution of $\hat{\theta}$.

c) Give the approximate asymptotic variance of $\hat{\theta}$.

Problem 4 — Asymptotic Normality of the MLE (Wasserman 9.9)

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, where σ^2 is known.

a) Derive the MLE $\hat{\mu}$.

b) Compute the Fisher Information for one observation.

c) Use the general MLE asymptotic normality theorem to derive the asymptotic distribution of $\hat{\mu}$.

d) Verify that this matches the exact finite-sample distribution.

Problem 5 — Hypothesis Testing Framework (C&B Ch 8)

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, where σ^2 is known.

We test:

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_1 : \mu \neq \mu_0.$$

a) Define the Type I and Type II errors in this context.

b) Derive a level-α test.

c) Compute the power function.

d) Show that the power increases with n. If power $\to 1$ as $n \to \infty$, a test is consistent. Is this test consistent?